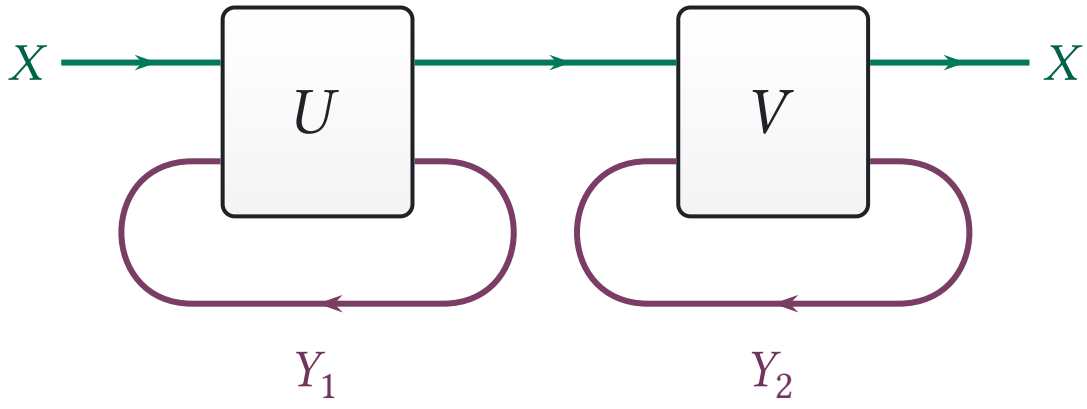
(a) two loops in sequence

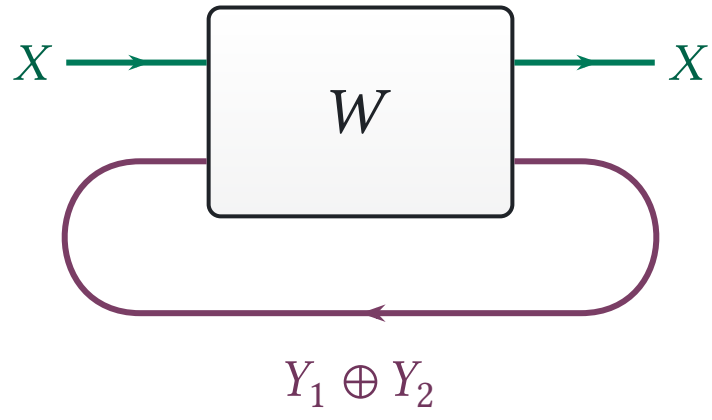
$$\mathrm{Tr}(W) = \mathrm{Tr}(V)\ \mathrm{Tr}(U)$$



$U : X \oplus Y_1 \longrightarrow X \oplus Y_1$

$V : X \oplus Y_2 \longrightarrow X \oplus Y_2$

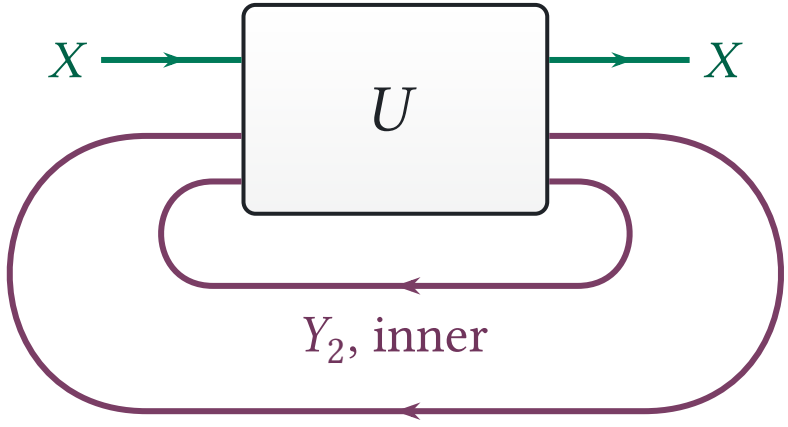
=



$W : X \oplus Y_1 \oplus Y_2 \longrightarrow X \oplus Y_1 \oplus Y_2$

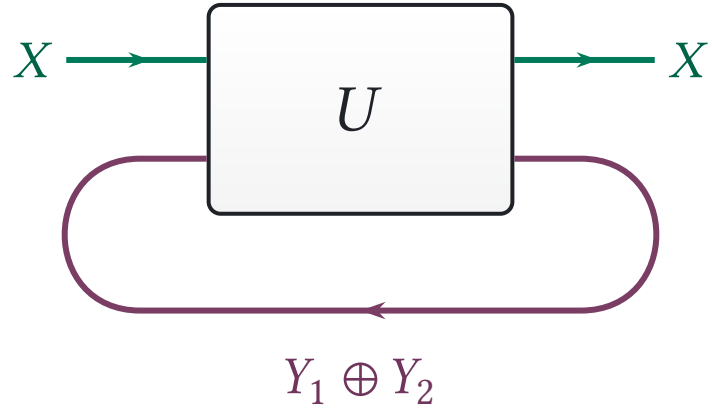
(b) one loop inside another

$$\mathrm{Tr}^{Y_1}\left(\mathrm{Tr}^{Y_2} U\right) = \mathrm{Tr}^{Y_1 \oplus Y_2} U$$



$U : X \oplus Y_1 \oplus Y_2 \longrightarrow X \oplus Y_1 \oplus Y_2$

=



$\mathrm{Tr}_{\oplus}^{Y_1 \oplus Y_2}(U) : X \longrightarrow X$